Swesan Pathmanathan

Canadian Citizen
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Education

McMaster University Hamilton, ON
Bachelor of Engineering, Software Engineering (Co-op) *Sep 2021 – Apr 2026*

Technical Skills

Languages: C, Python
DevOps & CI/CD: GitHub Actions, Docker
Systems: Linux, Windows, Bash
Backend: Node.js, REST APIs, PostgreSQL
Data: NumPy, scikit-learn

Work Experience

Advanced Micro Devices (AMD) | Python, Linux Markham, ON
Datacenter GPU Infrastructure & Validation Engineer (Co-op) *May 2024 – Aug 2025*

- Worked on post-silicon datacenter GPU system integration for MI300X, MI325X, and MI350X, running automated integration tests for performance and quality across 100+ test runs per week.
- Set up and ran tests on **Linux** and Windows GPU platforms with different BMC, BIOS, and OS images to check RAS, resets, IO, performance, and system management, cutting bring-up issues by about 20%.
- Wrote **Python** scripts and a workload launch tool using an internal framework to speed up how quickly engineers could run regression tests, reducing manual setup time from about 20 minutes to under 5 minutes.
- Helped debug post-silicon issues involving ROCm, power-management firmware, drivers, and hardware by reading logs, reproducing problems, and sharing clear findings with senior engineers, shortening time-to-root-cause on assigned issues by roughly 30%.
- Worked with infrastructure, firmware, and validation teams to track work in Jira and improve checklists and documentation, which made complex GPU tests easier to reproduce and reduced misconfigurations in new test setups.

Projects

ALU Design with Sequential Circuits | Verilog, Quartus *September 2025*

- Built an **Arithmetic Logic Unit (ALU)** in **Verilog** using simple, synthesizable modules, passing all planned functional tests.
- Used **Quartus** waveform views and testbenches to check that arithmetic and logic operations behaved as expected across dozens of input cases.

SufSort Sorting System | C, Bash, Linux *November 2025*

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.

Linux Log Corpus Analysis | Python, scikit-learn, NumPy *February 2026*

- Built a simple analysis pipeline in **Python** to read and process tens of thousands of labeled Linux system logs.
- Used **NumPy** and **scikit-learn** to find which log terms best separate different log categories, improving signal-to-noise for downstream analysis.

Relevant Courses

Real-Time Systems, Digital Systems Interfacing, Computer Architecture